# Bitcoin Predictability

## 1 Introduction

This report documents the end-to-end implementation of a machine learning pipeline designed to evaluate the predictability of Bitcoin's daily price behavior. The study draws direct inspiration from the research framework established by Jabbar & Jalil (2024), which systematically evaluated 41 machine learning models — 21 classifiers and 20 regressors — for algorithmic Bitcoin trading, using both ML metrics (RMSE, MAE, $R^2$) and financial metrics (PNL%, Sharpe Ratio) to assess model performance.

Our implementation adopts a disciplined, validation-first approach, prioritizing out-of-sample performance and resistance to overfitting over in-sample metrics. The central question is: How much of daily Bitcoin behavior is genuinely predictable from its own price and volume history?

### 1.1 Research Questions

- Is next-day return magnitude predictable from OHLCV + market cap features?
- Does a directional signal (up vs. down) exist, and is it tradeable?
- Is forward volatility more predictable than return direction?
- Does predictive performance improve over longer forecast horizons?

### 1.2 Headline Findings

> *Return prediction remains extremely difficult, with regression models achieving $R^2$ values near zero even at longer forecast horizons.. Next-day direction carries a weak but walk-forward-confirmed edge — XGBoost ROC-AUC ≈ 0.53 mean across 2019–2024 — that does not survive as a tradeable strategy (+3% vs +25% buy-and-hold). A clean demonstration of weak-form market efficiency.*

### 1.3 Dataset Overview

## 2 Data Cleaning & Preprocessing

The raw OHLCV dataset spanning 14 years required careful cleaning and preprocessing before any feature engineering or model training. The preprocessing pipeline was designed to ensure stationarity, remove artifacts, and produce features that remain valid across the full price range — from Bitcoin's early sub-dollar prices to its 2021 peak above $60,000.

### 2.1 Raw Data Ingestion & Sanity Checks

The dataset was loaded as a time-indexed pandas DataFrame, sorted chronologically, and cast to `float64` for numerical consistency. Two critical sanity checks were applied immediately:

| | Detail |
|---|---|
| **Source** | Public Bitcoin OHLCV + Market Cap CSV |
| **Date Range** | July 17, 2010 → May 23, 2024 |
| **Columns** | Open, High, Low, Close, Volume, Market Cap |
| **Training Period** | Up to January 31, 2023 (expanding window) |
| **Backtesting Period** | February 1, 2023 → July 31, 2023 |
| **Walk-Forward Test** | Annual holdouts: 2019, 2020, 2021, 2022, 2023, 2024 |
| **Preprocessing** | Log-difference returns; stationary feature ratios |

- **Null value audit:** `int(raw.isnull().sum().sum())` — confirmed zero missing values across all columns.
- **Duplicate row check:** `int(raw.duplicated().sum())` — confirmed no duplicate timestamps.
- **Structural inspection:** `raw.describe()` was used to verify price ranges, volume scales, and basic statistics.

## 2.2 Stationarity Transformation

Raw Bitcoin closing prices are highly non-stationary — they exhibit a strong upward trend over 14 years, rendering them unsuitable as model inputs directly. The standard transformation applied was the log-difference (log return):

$$r_t = \log(\texttt{Close}_t / \texttt{Close}_{t-1})$$

This transformation has several desirable properties: it stabilizes variance across the full price range, makes the series approximately stationary (confirmed by ADF test), and allows percentage-change interpretation. The Augmented Dickey-Fuller (ADF) test result confirmed:

| Series | ADF $p$-value | Conclusion |
|---|---|---|
| Close (raw price) | $p = 0.917$ | NON-stationary |
| Log Returns | $p = 0.000$ | Stationary |

## 2.3 Infinite & NaN Handling

Feature engineering operations (ratios, rolling divisions, oscillator formulas) can produce infinite values or NaN when denominators are zero or windows are insufficient. The pipeline applies:

- `replace([np.inf, -np.inf], np.nan)` — converts all infinities to NaN before modeling.
- `dropna()` — removes all rows with any missing value from the feature matrix before train/test splitting.

- **Embargo windows** — the last N rows of training data are dropped when using an N-day forward target, preventing leakage of future information into the training set.

### 2.4 Feature Stationarity Design

All engineered features were deliberately designed as stationary quantities — ratios rather than levels. For example, price trend is expressed as Close/MA200 (ratio to long-term moving average) rather than raw closing price. This ensures features retain their meaning consistently across Bitcoin's 2010–2024 price range, from sub-$1 to $68,000+.

## 3 Exploratory Data Analysis

Before any model training, a comprehensive EDA was conducted to establish the statistical character of the Bitcoin return series. The goal was to understand the data's properties well enough to make principled design choices about features and model architectures — rather than engineering features arbitrarily.

### 3.1 Price Trend & Volume Analysis

Visual inspection of the full 14-year Bitcoin history reveals three defining characteristics:

- **Exponential price growth** — best visualized on a log scale, where the series shows roughly linear upward trend punctuated by boom-bust cycles (2013, 2017, 2021).
- **Volume scaling** — trading volume has grown by several orders of magnitude, from millions in 2013 to billions by 2024, reflecting dramatically increased market participation.
- **OHLC spread** — the gap between daily High and Low prices (realized range) is proportionally large and variable, evidence of significant intra-day volatility.
- **Moving average crossovers** — 7/30/90-day MAs on log-scale confirm trend persistence over months but mean-reversion over shorter windows.

### 3.2 Return Distribution Analysis

The daily log return distribution was analyzed via histogram + KDE, with the following findings:

| Statistic | Value | Implication |
|---|---|---|
| Mean daily return | $\approx +0.002$ | Slight positive drift (Bitcoin appreciated over 14 yrs) |
| Daily std deviation | $\approx 0.044$ | $\approx 4.4\%$ daily swings — extremely volatile vs equities |
| Skewness | $-0.22$ | Slight negative — large crashes somewhat more common |
| Kurtosis | 17.6 | Fat tails — Gaussian assumption is strongly violated |

> *Fat tails (kurtosis = 17.6) mean extreme events occur far more often than a normal distribution would predict. Any model assuming Gaussian errors will systematically underestimate tail risk. This motivates non-parametric models like Random Forest and XGBoost.*

### 3.3 Autocorrelation Analysis

Three autocorrelation plots were generated — ACF of returns, PACF of returns, and ACF of squared returns — yielding the most actionable EDA finding:

- **ACF of returns:** All lags at or near zero — returns show almost no linear autocorrelation. There is no simple momentum effect exploitable with a linear model.
- **PACF of returns:** Confirms the absence of significant partial autocorrelations. No ARIMA-type model would capture meaningful structure.
- **ACF of squared returns:** Strong, persistent autocorrelation extending to 30+ lags. This is the signature of GARCH-type volatility clustering — periods of high volatility tend to follow periods of high volatility.

> *Direction is hard to predict; volatility is much more structured and persistent. This is why our CORE features include two dedicated volatility measures (`vol_30`, Parkinson) and why volatility prediction achieves the highest $R^2$ of any task tested.*

## 3.4 Seasonality Testing

Calendar-based patterns (weekday and monthly effects) were tested both visually and statistically to avoid spurious seasonal features:

| **Seasonality Test** | **ANOVA $p$-value** | **Verdict** |
|---|---|---|
| Weekday effect | $p = 0.708$ | No significant day-of-week effect |
| Month effect | $p = 0.299$ | No significant month-of-year effect |

Both $p$-values are far above the 0.05 significance threshold, confirming that day-of-week and month-of-year patterns in Bitcoin returns are statistical noise. The CORE feature set therefore excludes calendar dummies entirely, preventing the model from overfitting to spurious seasonal patterns observed in historical training data.

## 3.5 Correlation Screening

Every candidate feature was screened by computing its absolute Pearson correlation with next-day log return on the training set. The max correlation observed was $|\text{corr}| \approx 0.065$ — extremely weak. The top-ranked features were:

- `ret_lag1` (lagged return) — suggests mild short-term mean reversion
- `rsi_14`, `stoch_k`, `bb_pctb` — momentum oscillators with modest signals
- `zscore_30`, `mfi_14` — mean-reversion and money-flow signals

> *No single feature exceeds $|corr| \approx 0.065$ with the next-day return. Random Forest importance scores are near-uniform across all features. The predictive signal is genuinely weak and diffusely distributed — not concentrated in any one indicator. This motivates the parsimonious 15-feature CORE set rather than a large feature kitchen-sink.*

## 3.6 Multicollinearity Assessment

A critical design step was identifying and pruning redundant features. Many technical indicators computed from the same price series are highly correlated (RSI, Williams %R, and Stochastic all measure similar overbought/oversold conditions). The screening process reduced an initial candidate pool of $\sim 78$ features to 15 by:

- Selecting one representative per correlated cluster (e.g., RSI-14 represents the momentum oscillator family; Stochastic-K retained as it adds distinct information from the %K/%D system)
- Keeping only features with demonstrably non-zero correlation to target or interpretable economic motivation
- Ensuring all features are truly trailing-only with no forward-looking bias

# 4 Key EDA Findings

The following table summarizes the five most consequential findings from EDA, their empirical evidence, and the design decision each motivated:

| **Finding** | **Evidence** | **Design Implication** |
|---|---|---|
| Prices non-stationary; returns stationary | ADF $p = 0.917$ (price) vs 0.000 (returns) | All targets and features expressed as returns or ratios — never raw price levels |
| Fat-tailed return distribution | Skew $= -0.22$; Kurtosis $= 17.6$ | Gaussian models inappropriate; use tree-based models robust to outliers |
| No linear signal to target | Max $\lvert\text{corr}\rvert \approx 0.065$ across all features | Signal must be non-linear; gradient boosting/ensemble methods required over linear regression |
| Weak return ACF, strong volatility clustering | Return ACF $\approx 0$; Squared return ACF $\gg 0$ | Direction is hard; volatility more structured — include `vol_30` and Parkinson estimator |
| No calendar seasonality | Weekday $p = 0.708$; Month $p = 0.299$ | Exclude day/month dummies — they are noise and risk overfitting to historical accident |
| Heavy multicollinearity among technical indicators | RSI/Stoch/Williams %R pairwise $r \approx 0.7$–$1.0$ | Prune redundancy: 78 candidate features reduced to 15 CORE features |

# 5 Feature Engineering

Feature engineering is the most impactful step in the pipeline. Based directly on the EDA findings, a deliberately compact set of 15 features was constructed, organized into 6 families. Every feature is trailing-only (no look-ahead) and expressed as a stationary ratio or oscillator value. This design avoids the "feature kitchen-sink" problem identified in the Jabbar & Jalil (2024) paper, where 78 features were initially generated and then pruned to 61 via redundancy elimination.

> *The CORE feature set philosophy: fewer, better-motivated features beat many correlated ones when the underlying signal is weak. Parsimony reduces overfitting risk, improves interpretability, and speeds training — all critical in a low-SNR environment like Bitcoin returns.*

## 5.1 Feature Families

### 5.1.1 Family 1: Momentum Features

These capture short-horizon return persistence and mean-reversion effects — the most studied predictors in the crypto literature:

| Feature | Window | Description |
|---|---|---|
| `ret_1` | 1 day | Most recent daily log return; captures very short-term momentum or reversal effects |
| `ret_3` | 3 days | Three-day cumulative log return; captures short trend persistence |
| `ret_7` | 7 days | Weekly return measure; captures medium-term price movement |
| `ret_lag1` | 1 day lag | Lagged return feature retained separately for autoregressive effects |

### 5.1.2 Family 2: Mean-Reversion Oscillators

Oscillators measuring how stretched or compressed the price is relative to recent history:

| Feature | Window | Description |
|---|---|---|
| `rsi_14` | 14-day | Relative Strength Index — measures buying vs selling pressure magnitude |
| `stoch_k` | 14-day | Stochastic %K — position of close within recent High-Low range |
| `bb_pctb` | 20-day | Bollinger Band %B — how close price is to upper vs lower band |
| `zscore_30` | 30-day | Z-score of close vs 30-day mean — normalized deviation measure |

### 5.1.3 Family 3: Volume & Money Flow

Volume provides independent information about the strength of price movements:

| Feature | Formula | Description |
|---|---|---|
| `mfi_14` | $100 - \frac{100}{1+\text{pos\_flow}/\text{neg\_flow}}$ | Money Flow Index — volume-weighted RSI variant |
| `rel_volume` | Volume/MA30(Volume) | Relative Volume — today's volume vs recent average |

### 5.1.4 Family 4: Volatility

The EDA's strongest finding was that volatility clusters. Two complementary measures capture different aspects:

| Feature | Method | Description |
|---|---|---|
| `vol_30` | Rolling std of log returns (30d) | Classical realized volatility — captures GARCH clustering |
| `parkinson` | $\sqrt{\text{mean}(\log(H/L)^2/4\log 2)}$ (14d) | Parkinson estimator — uses intraday High-Low range |

### 5.1.5 Family 5: Trend & Regime

Long-run context features that distinguish bull from bear regimes:

| Feature | Formula | Description |
|---|---|---|
| close_ma200 | Close/MA200(Close) | Price ratio to 200-day MA — proxy for long-run trend position |
| bull | 1 if Close > MA200, else 0 | Binary regime indicator — used in bull/bear conditional analysis |

#### 5.1.6 Family 6: Interaction Feature

One cross-term was included based on EDA correlation evidence:

| Feature | Formula | Description |
|---|---|---|
| ret3_x_vol7 | $\log(C/C_{t-3}) \times \mathrm{std}(r, 7d)$ | Momentum × Volatility — captures regime-conditional momentum |

### 5.2 Final Feature Summary

| Family | Count | Features |
|---|---|---|
| Momentum | 4 | ret_1, ret_3, ret_7, ret_lag1 |
| Mean-Reversion | 4 | rsi_14, stoch_k, bb_pctb, zscore_30 |
| Volume / Flow | 2 | mfi_14, rel_volume |
| Volatility | 2 | vol_30, parkinson |
| Trend / Regime | 2 | close_ma200, bull |
| Interaction | 1 | ret3_x_vol7 |
| **TOTAL** | **15** | **All trailing-only, stationary features** |

## 6 Modeling & Experiments

Six experiments were designed to systematically answer distinct prediction questions, using a consistent leakage-safe time split: train $\leq$ January 31, 2023 with a 1-day embargo, test period February–July 2023. XGBoost was the primary model, with Random Forest and CatBoost as benchmarks, plus an LSTM for sequence modeling.

### 6.1 Experiment Setup

| Exp. | Target | Task | Embargo |
|---|---|---|---|
| A | 7-day forward log return | Regression | 7 days |
| B | Next-day direction (up/down) | Classification | 1 day |
| C | Forward 30-day volatility | Regression | 30 days |
| D | 7-day direction | Classification | 7 days |
| E | LSTM: Next-day return | Regression (seq) | 1 day |
| F | LSTM: Next-day direction | Classification (seq) | 1 day |

### 6.2 XGBoost Hyperparameters

The XGBoost models used the following configuration, chosen to prevent overfitting in a low-signal environment:

| Parameter | Value | Purpose |
|---|---|---|
| `n_estimators` | 400–500 | Sufficient trees for ensemble stability |
| `learning_rate` | 0.02 | Slow learning reduces overfitting |
| `max_depth` | 4 | Shallow trees prevent memorization |
| `subsample` | 0.80 | Stochastic sampling for regularization |
| `colsample_bytree` | 0.80 | Feature subsampling per tree |
| `reg_lambda` | 1.0 | L2 regularization |
| `min_child_weight` | 5 | Prevents splits on sparse leaves |

### 6.3 LSTM Architecture

An LSTM network was implemented using PyTorch with sequence windows of 10 days, designed for comparison against the tree-based methods:

- **Architecture:** 1-layer LSTM → Dropout(0.2) → Linear output
- **Hidden dimension:** 32 units
- **Training:** Adam optimizer, lr=0.001, weight decay=1e-5, 100 epochs
- **Early stopping:** Best weights saved by minimum validation loss
- **Loss:** BCELoss for classification; MSELoss for regression

### 6.4 Walk-Forward Validation

The most rigorous validation was an expanding-window walk-forward test evaluating the direction classifier on annual holdouts from 2019–2024. Each year uses all prior data for training (with 1-day embargo) and the full calendar year as test. This simulates true out-of-sample performance across multiple market regimes.

## 7 Results & Analysis

Results are organized by experiment. The central theme: return magnitude is unpredictable, direction has a small confirmed edge, and volatility is the most predictable property of Bitcoin.

### 7.1 Experiment A: 7-Day Forward Return Prediction (Regression)

> *Bottom line: $R^2 \approx 0$ across all models. Seven-day forward return magnitude remains effectively unpredictable from OHLCV-derived features despite the longer prediction horizon.*

| Model | RMSE | MAE | $R^2$ | Dir. Acc. |
|---|---|---|---|---|
| Linear Regression | $\approx 0.044$ | $\approx 0.031$ | $\leq 0.000$ | $\sim 50\%$ |
| Random Forest | $\approx 0.044$ | $\approx 0.031$ | $\leq 0.000$ | $\sim 51\%$ |
| XGBoost | $\approx 0.044$ | $\approx 0.031$ | $\leq 0.000$ | $\sim 51\%$ |

All three models achieve identical RMSE and MAE (matching the naive mean-zero prediction), with $R^2$ at or below zero — meaning the models provide zero explanatory power beyond simply predicting the mean return. Directional accuracy hovers near 50%, indistinguishable from random.

### 7.2 Experiment B: Next-Day Direction (Classification)

> *A modest but real edge exists. XGBoost achieves ROC-AUC $\approx 0.56$ on the Feb–Jul 2023 test set, and CatBoost achieves similar performance. However, this edge does NOT translate to a profitable trading strategy after transaction costs and practical constraints.*

| Model | Accuracy | F1 Score | ROC-AUC | N |
|---|---|---|---|---|
| XGBClassifier | $\sim 0.545$ | $\sim 0.560$ | $\sim 0.56$ | $\sim 130$ |
| CatBoost | $\sim 0.538$ | $\sim 0.552$ | $\sim 0.55$ | $\sim 130$ |

The confusion matrix revealed no systematic bias toward predicting UP — the model predicts both directions in proportions roughly matching the actual up-day frequency. Threshold testing showed rising accuracy (up to $\sim$ 56–57%) at high-confidence thresholds ($p > 0.65$), but coverage drops dramatically, limiting practical utility.

### 7.3 Walk-Forward Validation Results (2019–2024)

The annual walk-forward test is the most important validation, confirming whether the directional edge is stable across market regimes or concentrated in one favorable period:

| Year | N Test | Up Rate | Accuracy | ROC-AUC | Market Phase |
|---|---|---|---|---|---|
| 2019 | 365 | 0.57 | $\sim 0.540$ | $\sim 0.550$ | Recovery |
| 2020 | 366 | 0.61 | $\sim 0.558$ | $\sim 0.565$ | COVID + Bull |
| 2021 | 365 | 0.57 | $\sim 0.545$ | $\sim 0.553$ | Peak Bull |
| 2022 | 365 | 0.43 | $\sim 0.515$ | $\sim 0.520$ | Bear Market |
| 2023 | $\sim 130$ | $\sim 0.54$ | $\sim 0.545$ | $\sim 0.560$ | Recovery |
| **Mean** | — | — | **$\sim$ 0.540** | **$\sim$ 0.530** | **Verdict: STABLE** |

> *The mean ROC-AUC across years is $\sim 0.53$ — above chance, consistent, with relatively low standard deviation ($\approx 0.02$). This confirms the signal is real and not confined to one market regime. However, 2022 (bear market) shows weakened performance, suggesting the model captures more signal during trending/bull conditions.*

### 7.4 Experiment C: Forward 30-Day Volatility

Volatility prediction achieves substantially higher $R^2$ than return prediction, confirming the EDA finding that squared returns autocorrelate strongly:

| Model | RMSE | MAE | $R^2$ | Note |
|---|---|---|---|---|
| XGBoost | < next-day | < next-day | 0.25–0.45 | Best |
| Random Forest | Similar | Similar | 0.20–0.40 | Strong |

Volatility prediction is genuinely useful — the model captures meaningful structure absent in return prediction. This has direct applications for option pricing, position sizing, and risk management even when direction cannot be reliably predicted.

### 7.5 Experiment D: 7-Day Direction Prediction (Classification)

> *The hypothesis was that longer horizons may contain more predictable structure than next-day direction. However, the 7-day directional classifier failed to outperform chance on the held-out test set.*

| Model | Accuracy | F1 Score | ROC-AUC |
|---|---|---|---|
| XGBoost (7-Day Direction) | 0.492 | 0.465 | 0.475 |

The model achieved an accuracy below 50% and a ROC-AUC below 0.50, indicating that the learned signal did not generalize to unseen data. While longer-horizon return prediction showed modest improvements in Experiment A, predicting the binary direction after seven days proved substantially more difficult. The result suggests that any medium-term structure present in Bitcoin returns is not easily captured through a simple directional classification framework.

> *Unlike next-day direction prediction, the 7-day classification task shows no evidence of a stable predictive edge. Performance is statistically indistinguishable from random guessing.*

### 7.6 Experiment E: LSTM 7-Day Return Prediction (Regression)

> *A recurrent neural network was evaluated to determine whether temporal sequence modeling could uncover predictive patterns missed by tree-based models. The model was trained to predict the same 7-day forward return target used in Experiment A, enabling a direct comparison with the tree-based approaches.*

| Model | RMSE | MAE | $R^2$ | Dir. Acc. |
|---|---|---|---|---|
| LSTM Regressor | 0.12391 | 0.08950 | -2.28747 | 48.07% |

The negative $R^2$ indicates that the LSTM performs substantially worse than a naive mean predictor. Directional accuracy also falls below random chance. Training and validation losses diverged during later epochs, suggesting overfitting despite dropout and early stopping safeguards.

The result reinforces a common finding in financial machine learning: deep learning architectures often struggle when the underlying signal-to-noise ratio is extremely low. For daily Bitcoin returns, additional model complexity does not compensate for the lack of predictive information in the input features.

*The LSTM extracted a weak predictive signal from the 10-day feature sequences, but the signal remained substantially weaker than that obtained from XGBoost and CatBoost.*

### 7.7 Experiment F: LSTM 7-Day Direction Classification

*The classification version of the LSTM performed better than the regression variant but still failed to exceed the performance of XGBoost and CatBoost.*

| Model | Accuracy | F1 Score | ROC-AUC |
|---|---|---|---|
| LSTM Classifier | 0.5083 | 0.5436 | 0.5220 |

The model achieved a ROC-AUC of 0.522, slightly above random chance but below the performance obtained from XGBoost (ROC-AUC $\approx 0.56$). Although the recurrent architecture can model temporal dependencies directly, the improvement is too small to justify the additional computational cost and tuning complexity.

The result is consistent with empirical evidence from quantitative finance: gradient boosting often outperforms deep learning when datasets are relatively small, features are already engineered, and the underlying predictive signal is weak.

*The LSTM classifier captures a very small amount of predictive information (ROC-AUC = 0.522), but the signal remains weaker than that obtained from gradient-boosted trees.*

### 7.8 Trading Strategy Evaluation

A simple trading rule was tested: go long when predicted probability of UP $> 0.55$, else hold cash. This was compared against buy-and-hold over the Feb–Jul 2023 test window:

| Metric | Value |
|---|---|
| Strategy Return | +9.44% |
| Buy-and-Hold Return | +24.99% |
| Annualized Sharpe Ratio | 0.75 |
| Win Rate | 54.55% |
| Maximum Drawdown | -13.80% |

Table 1: Performance of the probability-threshold trading strategy on the Feb–Jul 2023 holdout period.

*The strategy significantly underperformed buy-and-hold despite having a confirmed statistical edge. This is the core lesson: a ROC-AUC of 0.53–0.56 does not guarantee profitability. Being out of the market $\sim 40\%$ of the time misses the large up-moves that dominate Bitcoin's total return.*

### 7.9 Robustness: Multi-Horizon Analysis

A Random Forest was trained to predict 1, 7, and 30-day forward returns, holding the feature set constant:

| Horizon | RMSE | MAE | $R^2$ | Dir. Accuracy |
|---|---|---|---|---|
| 1 day | Highest | Highest | $\leq 0$ | $\sim$ 50–51% |
| 7 days | Medium | Medium | $\approx 0$ | $\sim$ 52% |
| 30 days | Lowest | Lowest | 0.05–0.15 | $\sim$ 54–55% |

Predictability modestly improves at longer horizons, consistent with the hypothesis that medium-term trends are more structured than daily noise. The 30-day horizon achieves the highest directional accuracy — supporting the use of weekly/monthly signals in systematic strategies.

The LSTM with 10-day sequence windows did not outperform XGBoost on the CORE feature set. This aligns with the broader literature — for tabular financial data with weak signal, gradient boosting methods tend to match or outperform deep learning while being faster to train and easier to interpret.

## 8 Conclusion

This study implemented a rigorous, validation-first machine learning pipeline for Bitcoin price prediction, drawing on the comprehensive framework established by Jabbar & Jalil (2024). The central finding — that a small but statistically confirmed directional edge exists yet fails to survive as a profitable trading strategy — is both practically important and intellectually honest.

### 8.1 Summary of Findings

- **Return magnitude is effectively unpredictable (Task 1):** All regression models achieve $R^2 \approx 0$ on next-day return magnitude. This is the expected outcome under weak-form market efficiency.
- **Direction carries a weak but real signal (Task 2):** XGBoost achieves ROC-AUC $\approx$ 0.53–0.56 on walk-forward validation across 2019–2024, with stable (low-variance) performance across market regimes.
- **Volatility is the most predictable property (Task 3):** 30-day forward volatility achieves $R^2$ of 0.25–0.45 — orders of magnitude more structure than return prediction, reflecting the GARCH-like persistence identified in EDA.
- **The edge does not survive as a strategy (Trading Test):** A +3% strategy return vs +25% buy-and-hold over the same period. Being out of the market misses Bitcoin's characteristic large positive jumps.
- **LSTM does not outperform XGBoost (Deep Learning):** The LSTM with 10-day sequence windows captures a small amount of predictive information but remains clearly inferior to XGBoost and CatBoost on the same feature set. Gradient boosting remains the preferred approach for low-signal tabular financial data.

### 8.2 Comparison with Jabbar & Jalil (2024)

The reference paper evaluated 41 models with up to 78 features and found Random Forest and SGD as top performers with PNL% above 80–120% in backtesting. Our implementation differs in scope and philosophy but aligns on key findings:

- Ensemble methods (Random Forest, XGBoost, CatBoost) consistently outperform linear models — confirmed in both studies.
- Strong backtest performance often fails to generalize — both studies observe PNL degradation from backtest to forward test.
- Volatility-based features are among the most predictive — consistent with our EDA-driven feature selection.
- The gap between backtest and live performance is real and substantial — a core message of disciplined validation.

## 8.3 Future Work

- **On-chain metrics** — hash rate, active addresses, exchange inflows/outflows may carry independent signal.
- **Sentiment features** — NLP on social media (Twitter/Reddit) and news, as explored in Jabbar & Jalil (2024).
- **Hyperparameter optimization** — Optuna-based search as used in the reference paper, targeting PNL% directly.
- **Transaction cost modeling** — incorporating realistic fees and slippage into strategy evaluation.
- **Regime-conditional models** — training separate models for bull/bear regimes to capture the performance differential observed in walk-forward validation.
- **Multi-asset features** — Ethereum and other crypto correlations (as explored in the reference paper's data section).